

Curriculum Vitae

Sagnik Sen

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I am a research scientist developing innovative sequencing methods at the intersection of epigenomics, spatial imaging, and machine learning. My research focuses on creating new approaches to interrogate chromatin alterations, ranging from sequencing-based assays such as NEED-seq and the FFPE-compatible FERN-seq to spatial, imaging-based chromatin profiling technologies. I also develop AI/ML and network-based analytical frameworks to interpret these complex datasets and apply them to fundamental biological questions, including how chromatin organization and DNA modification states regulate cell fate decisions during development, influence responses to epigenetic therapies, and shape the aging brain. Working seamlessly across both experimental and computational disciplines, I am currently developing an independent research program focused on chromatin reorganization during cell maturation. I am passionate about interdisciplinary collaboration and committed to mentoring and supporting junior scientists.

Career Highlights

- Ten years' experience as Bioinformatic Researcher
- Gold Medalist in Master's (M.Tech in Bioinformatics)
- Journals: 26(peer-reviewed), Book Chapters: 1, Conference/Workshop papers/Poster Presented: 7, Journals Under Review: 5, Under Preparation: 2
- Enthusiastic Researcher working at the confluence of Computer Science and Biology

Academics

1. Ph.D. in *Engineering*, Jadavpur University, India, 2016-2021.
2. Master of Technology in *Bioinformatics*, SRM University, India, 2015 (1st class, Hons., 1st Rank).
3. Bachelor of Technology in *Computer Science and Engineering* from Haldia Institute of Technology, India, 2013 (1st class, Hons.).

Areas of Research Interest

- **Technology Research & Development**- High throughput Sequencing/ Epigenetic Mechanism/ Single cell genomics and epigenomics/ Transcriptomics/ Multi-Omic Study/ Chromosome Architecture

-**Algorithm Design**- Graph Theoretical Modelling/ Computational Biology/ Computational Structural Biology/ Machine Learning/ Statistical Analysis/ Mathematical Modeling/ Graph Neural Network/ Transformers/ Data augmentation

-**Basic & Translation Research**- Epigenetic Target driven mechanism/ Aging/ Drug repurposing/ Epigenetic diagnostic Markers Study

Experience in Assays:

- NEED-seq/NicE-seq/EM-seq
- FERN-seq (FFPE Epitope-Restricted Nick-translation Sequencing, an advanced NEED-seq protocol for archival tissue)
- Western Blots

Experience in Data Analysis Pipeline Design and Implementation:

NEED-seq, ChIP-seq, NicE-seq, NiceView-seq (Image segmentation study), ATAC-seq, EM-seq, WGBS, Hi-C, RNA-seq, scRNA seq, Histopathology Image data analysis

Pipeline Highlights

- Automated End-to-end pipeline for NEED-seq upstream analysis
- Down-stream pipeline for in-depth study on epigenetic markers, Transcription factors, and DNA-methylation from high throughput sequencing (such as, NEED-seq, ChIP-seq, NicE-seq etc)
- Automated fluorescent image analysis based pipeline to study open chromatin landscape from NicE-View-seq microscopic Images
- Pipeline based on graph theoretical modelling to study the impact of transcription factors on regulation of oncogenesis and tumour suppression considering comparative Multi-Omics environment.
- Pipeline based on graph theoretical modelling and mathematical modelling (numerical methods) to study mechanism of therapeutic agents (e.g., CX5461) within Multi-Omics environment (including epigenomics, chromosome architecture)
- Pipeline to analyze Hi-C data
- Pipeline for comparative study between DNA methylation data and Transcription factors (Specially for ChIP-BS-seq/ TEM-seq(yet to publish))
- Pipelines based on network architecture, unsupervised learning and mathematical modelling for studying structural aberrations of Intrinsically Disorder Protein depending on evolutionary sequence space and structure space analysis
- Artificial Intelligence (based on Convolutional Neural Network) based Histopathology Image analysis model for prediction of breast tumour subclasses.

Proficient Computational Languages/Pipelines

- Python, R, Bash
- Nextflow, Snakemake

Working Environment

- Anaconda, VS Code, HPC, Windows, Unix, Mac OS

Work and Research Experience

- Collaborators, Prof. Petra Hajkova, Professor of Developmental Epigenetics, Imperial College of London, Since 2025
- **Research Scientist I, Molecular Genetics and Genomics Division, New England Biolabs, Since February, 2024.**
- Project linked Person, SBIR Collaborative Project with Epiccypher Inc, since 2022.
- Collaborators, Dr. Brad Langhorst, Dr. Louise Williams and Dr. Chaithanya Ponnaluri, New England Biolabs, Since 2022.
- Collaborator, Prof. Laura Landweber, Center for Computational Biology and Bioinformatics, Columbia University, Since 2022.
- Collaborator, Prof. Melanie Ehrlich, Tulane Cancer Center, Tulane University, Since 2021.
- **Post-Doctoral Scientist, Genome Biology Division, New England Biolabs Inc, Since September, 2021-Feb 2024**
- Research Fellow, Jadavpur University, India, Since June 2015.
- Collaborative Researcher, CSIR-Indian Institute of Chemical Biology, India, Since November 2016.
- Collaborator, Prof. Vladimir Uversky, Molecular Medicine, University of South Florida, Tampa, Since 2017.
- Collaborator, Dr. Amit Sharma, University Hospital Bonn, Germany, Since March 2019.
- Collaborator, Eischen Laboratory, Thomas Jefferson University, Philadelphia, Since February 2019.
- Master's Thesis Joint Supervision: 10 students finished

Panel Discussion:

- Invited Talk on "Exploring DNA-Seq Techniques And Impact Of Epigenetics" at IEEE YIP Kolkata Chapter (Jadavpur University), Kolkata, India, Dec, 2022.
- Panel Discussion on "Careers in Industry (Research)", Behavioral Plasticity Research Institute, Texas, USA, Apr, 2025.

Academic Activities

- Topic Editor, Frontiers in Bioinformatics
- Reviewer of Current Opinion of Structural Biology
- Reviewer of NAR Genomics and Bioinformatics
- Review Editor, Computational Genomics, Frontiers in Genetics
- PC member of Function COSI: Protein Function Annotation, ISMB/ECCB 2019.
- Reviewer of SADHANA
- Reviewer of Briefings in Bioinformatics
- Reviewer of Nature Communication Biology
- Reviewer of Life, MDPI
- Reviewer of Cancer, MDPI
- Reviewer of Ecotoxicology and Environmental Safety
- Reviewer of IEEE-TCBB

Awards/Fellowships

1. Nominated for 2026 BBA Rising Stars in Biochemistry award.
2. Emerging Scientist Award in 11th International Scientist Awards on Engineering, Science and Medicine by VDGGOOD Technology.
3. DST-Inspire Fellowship from Department Science and Technology INSPIRE scheme, India, 2016-2021.
4. Gold Medallist from Department of Bioinformatics during Master's Program, SRM University, India, 2015.

Publication Citation

459 citations have been received as per the information in Google Scholar with an *h*-index of 11 and i10-index 15.

List of Publications

+ Corresponding Author

Journal (published/accepted)

*First Author/Joint First Author

1. A. Dey, D. Sanyal D, K. Chattopadhyay, U. Maulik, V. Uversky and S. Sen+ “IMMUND: A Diagnostic and Therapeutic Pipeline to Uncover the Convergence in Functional Perturbation at Early Stages of Neurodegenerative Diseases and Multiple Sclerosis Based on Protein Markers. *Int. J. Mol. Sci.* 2026, 27, 5627. <https://doi.org/10.3390/ijms27125627>
2. S. Sen*, et al. (2025) “Distinct structural and functional heterochromatin partitioning of lamin B1 and lamin B2 revealed using genome-wide nicking enzyme epitope targeted DNA sequencing”, *Nucleic Acids Research* 53(2): gkae1317.
3. S. Sen* et al. "The myoblast methylome: multiple types of associations with chromatin and transcription." *Epigenetics* 20.1 (2025): 2508251.
4. M. Ehrlich, K. C. Ehrlich, M. Lacey, C. Baribault, S. Sen, P. O. Estève, and Pradhan, S. “Epigenetics of genes preferentially expressed in dissimilar cell populations: Myoblasts and cerebellum”, *Epigenomes*, 8(1), 4, (2024).
5. VUN Sunitha Kumary, B. J. Venters, Raman K., S. Sen, et al. “Emerging Approaches to Profile Accessible Chromatin from Formalin-Fixed Paraffin-Embedded Sections”, *Epigenomes* 8.2 (2024): 20.
6. P. O. Estève, S. Sen*, V. US, et al. “Poly ADP-ribosylation of SET8 leads to aberrant H4K20 methylation domains in mammalian cells”, *Communications Biology*. 2022 Nov 25;5(1):1292.
7. K. C. Ehrlich, M. Lacey, C. Baribault, S. Sen, et al. “Promoter-Adjacent DNA Hypermethylation Can Downmodulate Gene Expression: TBX15 in the Muscle Lineage”, *Epigenomes* 2022, 6, 43.
8. R. Das, U. B. Maulik, B. Boote, S. Sen, et al. “Multi-path Convolutional Neural Network to identify Tumorous Sub-classes for Breast Tissue from Histopathological Images.” *SN Computer Science*, 3(5), 1-11, (2022).
9. A. Dey, S. Sen+, et al. “Study of transcription factor druggability for prostate cancer using structure information, gene regulatory networks and protein moonlighting.” *Briefings in Bioinformatics*, 23(1), bbab465, 2022.
10. S. Sen*, A. Dey, et al. “Studying the effect of alpha-synuclein and Parkinson's disease linked mutants on inter pathway connectivities”, *Sci Rep.* 2021 Aug 11;11(1):16365. doi: 10.1038/s41598-021-95889-5. PMID: 34381149; PMCID: PMC8358055.
11. S. Sen*, A. Dey, S. Bandyopadhyay, U. Maulik, and V. Uversky, “Understanding Structural Malleability of the SARS-CoV-2 Proteins and their Relation to the Comorbidities”, *Briefings in Bioinformatics*, Volume 22, Issue 6, November 2021, bbab232.
12. A. Dey, S. Sen+, and U. Maulik “Unveiling COVID-19-associated organ-specific cell types and cell-specific pathway cascade”, *Briefings in bioinformatics*. 2021 Mar;22(2):914-23.
13. S. Begum, R. Sarkar, D. Chakraborty, S. Sen, and U. Maulik, “Application of active learning in DNA microarray data for cancerous gene identification”, *Expert Systems with Applications*, 2021, 177, p.114914.
14. A. Dey, S. Sen+, V. N. Uversky and U. Maulik “Structural Facets of POU2F1 in Light of the Functional Annotations and Sequence-Structure Patterns”, *Journal of Biomolecular Structure and Dynamics*, 2021, DOI: 10.1080/07391102.2020.1733092.
15. M. Ghosh, S. Sen, R. Sarkar, and U. Maulik “Quantum squirrel inspired algorithm for gene selection in methylation and expression data of prostate cancer”, *Applied soft computing*. 2021 Jul 1;105:107221.
16. R. Das, S. Sen+, and U. Maulik, “A Survey on Fuzzy Deep Neural Networks”, *ACM Computing Survey*, 53(3), pp.1-25, 2020.

17. K. K. Ghosh, S. Ghosh, S. Sen, R. Sarkar, and U. Maulik, (2020). "A two-stage approach towards protein secondary structure classification", *Medical Biological Engineering Computing*, 2020.
18. A. Sharma, A. Biswas, H. Liu, S. Sen, A. Paruchuri, P. Katsonis, O. Lichtarge, T. C. Dakal, U. Maulik, M. M. Gromiha, S. Bandyopadhyay, M. Ludwig, F. G. Holz, K. U. Loeffler and M. C. Herwig-Carl, "Mutational landscape of the BAP1 locus reveals an intrinsic control to regulate the miRNA network and the binding of protein complexes in Uveal Melanoma", *Cancers*, 11(10), p.1600, 2019.
19. S. Chowdhury, D. Sanyal, S. Sen*, V. N. Uversky, U. Maulik, and K. Chattopadhyay "Evolutionary analyses of sequence and structure space unravel the structural facets of SOD1", *Biomolecules*. 2019 Dec 4;9(12):826.
20. S. Chowdhury, S. Sen*, A. Banerjee, V. N. Uversky, U. Maulik, and K. Chattopadhyay, "Network mapping of the conformational heterogeneity of SOD1 by deploying statistical cluster analysis of FTIR spectra", *Cellular and Molecular Life Sciences*, PP1-10, 2019.
21. S. Sen*, A. Dey, S. Chowdhury, K. Chattopadhyay, U. Maulik, "Understanding the Evolutionary Trend of Intrinsic Structural Disorders in Cancer Relevant Proteins as probed by Shannon Entropy Scoring and Structure Network Analysis", *BMC Bioinformatics* pp. 231-242, 19(13), 2019.
22. A. Sannigrahi, D. Mullick, D. Sanyal, S. Sen, U. Maulik, K. Chattopadhyay, "Effect of Ergosterol on the Binding of KMP-11 with Phospholipid Membranes: Implications in Leishmaniasis", *American Chemical Society Omega*, 4(3), pp. 5155-5164, 2019.
23. S. Sen*, and U. Maulik, "Recent advancement toward significant association between disordered transcripts and virus-infected diseases: a survey". *Briefings in functional genomics*, pp. 458-470 17 (6), 2018.
24. U. Maulik, S. Sen, S. Maulik and S. Bandyopadhyay, "Detecting TF-MiRNA-Gene Network Based Modules for 5hmC and 5mC Brain Samples: A Intra- and Inter-Species Case-Study Between Human and Rhesus", *BMC genetics* 19(9), 2018.
25. U. Maulik, V. Uversky and S. Sen+, "A statistical approach to Detect Intrinsically Disordered Proteins associated with Uterine Leiomyoma", *Protein Peptide Letters*, 25(5), pp. 483-491, 2018.
26. S. Mallik, S. Sen and U. Maulik, "IDPT: Insights into Potential Intrinsically Disordered Proteins Through Transcriptomic Analysis of Genes for Prostate Carcinoma Epigenetic Data", *Gene*, vol. 586, pp. 87-96, 2016.

Journal (Under Review)

1. S. Sen*, P. Esteve, et al. Altered chromatin accessibility and nucleosome positioning landscape upon HDAC and LSD1 inhibition in cancer cell (under review in *Science Advances*).
2. S.Sen* et al. Developmental Trajectory of Epigenetic Targeted Drugs : A Review. (under Review in *BBA- Gene Regulatory Mechanisms*).
3. Angelova MT, Feng Y, Villano DJ, Aslan E, Lu MW, Estève PO, Sen S, Jack WE, Pradhan S, Landweber LF. A PIWI protein-dependent DNA N6-adenine methylation pathway in *Oxytricha* protects genomic sequences from deletion. *bioRxiv*. 2026 Jan 7:2026-01 (under review in *Gene & Development*).
4. Erickson, J et al., "Targeting genome maintenance in glioblastoma with a histone deacetylase inhibitor and DNA break-mimetic AsiDNA." (Under Review, *NAR Cancer*)
5. D. Evanich, L. N. Blum, S. Sen*, N. Dai, B. W. Langhorst, M. A. Campbell, L. Williams, V. K. C. Ponnaluri, S. Pradhan, "Dynamic DNA methylation and hydroxymethylation regulate transcription during neuronal cell differentiation." (Under Review, *Genome Biology*)

Arxiv/BioRxiv (in preparation)

1. Estève PO, Sen S*, Raman K, Unnikrishnan A, Pradhan S. NPM1 mislocalization mediated by RNA Pol I inhibition alters chromatin landscape. *bioRxiv*. 2025 Oct 28:2025-10.

2. S. Sen*, A. Chatterjee, and U. Maulik. "ImmunAL: a frame to identify the immunological markers for Mild Moderated Alzheimer's Disease applying Multiplex Network Model." bioRxiv (2021): 2021-10

Patent

1. S. Sen*, P. O. Esteve et al. "Compositions and methods for detecting Damaged molecular Targets on chromosomal DNA", Application No.-64/098,797, Receipt Date-06/25/2026.

Invited as Speaker

1. "Pan-cancer genome-wide chromatin profiling on archival tumor tissue unveiling transcriptional landscape", Advancing the Epigenome: Emerging Tools and Technologies,, **DNA, RNA, and Chromatin Epigenetics: Disease and Development meeting, October 15-17, 2025 at Northwestern University's Simpson Querrey Biomedical Research Centre (Chicago), 2025 Epigenetics Society International Conference.**

Conference/Workshop Proceedings/Poster Presentations

1. S. Sen, D. Tarasia et al. "NICKL: A deep learning pipeline for measuring Chromatin Accessibility from NicE-view-Seq images" Biological Data Science 2024, Cold Spring Harbor Laboratory (Accepted).
2. S. Sen, P. Esteve, et al. "Poly ADP-ribosylation of SET8 leads to aberrant H4K20 methylation in mammalian nuclear genome", XXIII International Congress of Genetics, 2023 (Accepted).
3. S. Sen, A. Dey and U. Maulik, "Unravelling the Dynamicity of POU2F1 based on Evolutionary Conservation and Structure Network Analysis", Function COSI, ISMB/ECCB 2019 (Accepted).
4. A. Dey, R. Das, S. Sen and U. Maulik, "moonPRED: A Deep Neural Network Based Tool to Predict Protein Moonlighting moonPred, MLCSB COSI ISMB/ECCB, 2019 (Accepted).
5. S. Sen, A. Dey and U. Maulik, "Identifying Transcription Factors as Drug Targets for Prostate Carcinoma Considering Protein Moonlighting Behavior", RECOMB 2019 (accepted).
6. S. Sen, A. Dey and U. Maulik, "Identifying Potential Hubs for Kidney Renal Clear Cell Carcinoma from TF-miRNA-Gene Regulatory Networks", IEEE ASPCON.(Accepted)
7. S. Sen, and U. Maulik, "Detecting the Common Biomarkers for Early Stage Parkinson's Disease and Early Stage Alzheimer's Disease Associated with Intrinsically Disordered Protein", International Conference on Advances in Computing and Data Sciences, pp. 25-34, 2018.

Book Chapter

1. S. Sen*, R. Das, S. Dasgupta and U. Maulik, " Application of Deep Architecture in Bioinformatics", Deep Learning Techniques For Biomedical And Health Informatics, Springer. (Accepted)

Google Scholar link:

https://scholar.google.com/citations?hl=en&user=G76NdKAAAAAJ&view_op=list_works&sortby=pubdate

Github Link:

<https://github.com/Sagnik-Epigen>

LinkedIn:

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Date: 09/19/2026

